

Estimating QSVT angles for matrix inversion with large condition numbers

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ABSTRACT

Quantum Singular Value Transformation (QSVT) is a state-of-the-art, near-optimal quantum algorithm that can be used for matrix inversion. The QSVT circuit is parameterized by a sequence of angles that must be pre-calculated classically, with the number of angles increasing as the matrix condition number grows. Computing QSVT angles for ill-conditioned problems is a numerically challenging task. We propose a numerical technique for estimating QSVT angles for large condition numbers. This technique allows one to avoid expensive numerical computations of QSVT angles and to emulate QSVT circuits for solving ill-conditioned problems.

1. Introduction

1.1. Motivation

Quantum Singular Value Transformation (QSVT) [1–3] is a modern quantum algorithm used to perform Hamiltonian simulation, solve systems of linear equations, and, in general, compute polynomials of matrix singular values. The shape of a polynomial computed by QSVT is determined by a sequence of angles ϕ that must be pre-calculated classically and parameterize the QSVT circuit. In particular, the QSVT algorithm is a near-optimal method for matrix inversion and can, therefore, serve as a quantum linear solver. In this context, the number of QSVT angles grows at least linearly with the matrix condition number ρ . However, many systems of practical interest are stiff, with corresponding matrices characterized by high ρ . In such cases, high-precision computation of ϕ [4–6] becomes a challenging numerical task already for ρ around 10^3 . Notably, the computational cost of these methods [4,5] grows quadratically with ρ . The complexity was reduced to near-linear using the so-called Fast-Fixed-Point-Iteration (FFPI) technique recently proposed in Ref. [7], which significantly accelerates the high-precision computation of the QSVT angles. In our work, we propose an alternative fast numerical technique to estimate QSVT angles for large condition numbers. This approach allows QSVT to be applied to approximately solve ill-conditioned systems of linear equations, while significantly reducing the numerical resources required for computing QSVT angles.

The estimation algorithm proposed in this work was motivated by the following question: is it possible to reduce the optimization of a quantum circuit that depends on N_{pars} parameters to an optimization with respect to $N_{\text{meta}} \ll N_{\text{pars}}$ metaparameters? Let us assume that we have a quantum circuit $C[p]$ computing an object function $f(x)$ depending on the variable x , and this circuit depends on N_{pars} parameters $p(x)$. Then, the direct approach for constructing $C[p]$ for a particular x is the minimization of the cost function

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Table 1Summary of the numerical technique for computing the estimated QSVT angles $\bar{\phi}$ for the target value κ_0 .

<i>Computing metaparameters from the reference high-precision QSVT angles.</i>	
I.1	Compute N_{ref} sets of high-precision QSVT angles for various $\kappa_{\text{ref},j}$ with $j = 0, 1, \dots, (N_{\text{ref}} - 1)$ by using the minimization (17).
I.2	Find $c_{\text{ampl},j}$ using Eq. (24).
I.3	Choose a single set of high-precision QSVT angles, ϕ_{ref} , for some κ_{ref} . It is better to take ϕ_{ref} for $\kappa_{\text{ref}} = \max_j \kappa_{\text{ref},j}$.
I.4	Compute $c_{\text{sh},j}^{\text{pos}}$ and $c_{\text{sh},j}^{\text{neg}}$ using Eq. (29).
<i>Estimating QSVT angles using the calculated metaparameters.</i>	
II.1	Choose the target value κ_0 .
II.2	Compute N_a for the chosen κ_0 using Eqs. (22).
II.3	Find the positive and negative envelopes of the estimated angles by using Eq. (28). Mirror the copies of the envelopes and concatenate them with the original ones to obtain the full envelope.
II.4	Find the maximum value of the estimated shifted angles using Eq. (23).
II.5	Compute the values of the estimated QSVT angles $\bar{\phi}$ by using Eqs. (30) and (19).

$L(p) = |f - C[p]|$ to find the most optimal parameters p . Instead, one can try to find $N_{\text{meta}} \ll N_{\text{pars}}$ metaparameters m that capture the dependence of p on x . These metaparameters can thus be considered a compressed version of the original parameters p . In this case, minimizing the new cost function $L(m) = |f - C[m]|$ should require far fewer numerical resources than minimizing $L(p)$. We find such metaparameters for the QSVT circuit in the matrix-inversion problem and, by using them, estimate the QSVT angles. However, it is important to emphasize that the proposed technique is based solely on observations of the specific behavior of the QSVT angles for matrix inversion, and its rigorous analytical justification remains an intriguing open question.

1.2. Applications of quantum linear system algorithms (QLSAs)

Quantum Linear System Algorithms (QLSAs), such as QSVT, are used to solve a broad range of linear and nonlinear (NL) problems. In particular, QSVT was used to simulate stationary linear electromagnetic (EM) waves [8] and was proposed for modeling kinetic waves in an electron plasma [9]. It was also proposed in [10] to use preconditioned QLSAs for the linear hydrological modeling, where it was shown that condition numbers of practical interest can reach 10^6 or even larger values. Similarly, it was described in [11,12] how QLSAs can be applied to modeling linear geological flows. In general, QLSAs are also used to solve time-dependent differential equations [13].

On the other hand, NL problems linearized using the Carleman [14–16] or Koopman–von Neumann [17–19] techniques can be simulated with QLSAs. For instance, QLSA is a part of the Carleman-based algorithm discussed in [15] for modeling dissipative dynamics with quadratic nonlinearity. The same approach was proposed in [20] for solving first-order differential equations used in chemical kinetics.

Thus, QLSAs encompass a broad class of algorithms widely used for modeling both linear and NL dynamics, with QSVT being a well-known and versatile QLSA.

1.3. Notation

Here, we summarize the main notation used throughout this work. The scalar κ_{qsvt} is a parameter that defines an accurate approximation of the inverse function [Eq. (7)]. It is also an input parameter for the minimization algorithm [4] described in Sec. 2.2, which computes high-precision QSVT angles ϕ . The scalar κ_{ref} is a specific value of κ_{qsvt} used to compute high-precision QSVT angles ϕ_{ref} , which serve as a reference dataset for the estimation technique presented in Sec. 3. The parameter κ_0 is the value of κ_{qsvt} used by the estimation algorithm to compute the estimated QSVT angles, denoted as $\bar{\phi}$. The scalar ρ_A is the condition number of the target matrix A that needs to be inverted.

1.4. Key results

We propose a numerical technique for estimating QSVT angles for inverting matrices characterized by large condition numbers. The computational time of this technique scales linearly with κ_0 . The algorithm is summarized in Table 1. In particular, a block-encoded matrix A can be inverted by the QSVT circuit with high precision using the QSVT angles ϕ computed by the minimization algorithm [4] with $\kappa_{\text{qsvt}} \geq \|A\|^{-1} \rho_A$, where $\|A\|$ is the spectral norm of A . The maximum error ϵ_{qsvt} of these QSVT computations is determined by the chosen degree N_c of the polynomial (12a) approximating the inverse function (7). Similarly, the matrix A can be inverted by the QSVT circuit using the estimated angles $\bar{\phi}$ computed for $\kappa_0 \geq \|A\|^{-1} \rho_A$ with the estimation algorithm described in Sec. 3. Assuming the result of the inversion is normalized to one, we find that the maximum error ϵ_{appr} of the QSVT computations with the angles $\bar{\phi}$ gradually increases with κ_0 and is approximately 10^{-5} for $\kappa_0 \leq 10^6$ for matrices with $\|A\| < 1$.

This paper is organized as follows. In Sec. 2, we review the QSVT algorithm for matrix inversion and the minimization approach for computing QSVT angles with high precision. In Sec. 3, we describe the numerical technique to estimate QSVT angles that can be

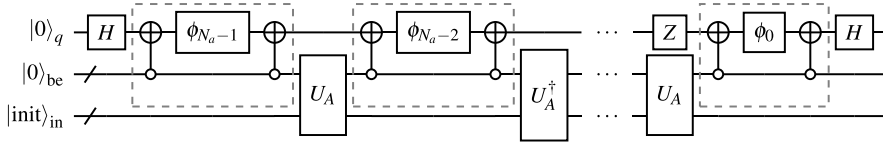


Fig. 1. The QSVT circuit computing $P[\phi](A)$ [Eq. (3)]. Here, the gates schematically denoted as ϕ_j represent the rotations $R_z(2\phi_j)$.

used for inverting matrices with large condition numbers. In Sec. 4, we test the estimated angles on various examples including a linear boundary-value problem of stationary EM waves. In Sec. 5, we present the main conclusions.

2. Overview of QSVT for matrix inversion

2.1. Overview

Let us assume that we have an invertible $N \times N$ matrix A whose singular value decomposition is

$$A = U_l S U_r^\dagger, \quad (1)$$

where U_l and U_r are unitary matrices containing left and right singular vectors of A , correspondingly, and the matrix $S = \text{diag}(s_0, \dots, s_{N-1})$ contains real strictly positive singular values s_i of A . QSVT [1,2] builds a function $P(A)$ by transforming the singular values s_i :

$$P(A) = U_l f(S) U_r^\dagger, \quad (2)$$

where $f(S) = \text{diag}(f(s_0), \dots, f(s_{N-1}))$, and it is assumed that $\|A\| \leq 1$, i.e. $s_{\max} \leq 1$. (Further in the text, we use the small letter s without any subindex to indicate either some singular value, or a set of singular values of A .) In particular, the QSVT calculation of odd functions can be performed using the circuit shown in Fig. 1. It consists of a sequence of $N_c + 1$ parameterized rotations $\exp(i\phi_k Z_\Pi)$ (dashed gray boxes in Fig. 1), alternating with N_c calls to the oracle U_A , which block-encodes the target matrix A :

$$P[\phi](A) = \langle + |_q \langle 0 |_{\text{be}} \left(e^{i\phi_0 Z_\Pi} Z_q U_A e^{i\phi_1 Z_\Pi} \prod_{k=1}^{(N_c-1)/2} G_k \right) | + \rangle_q | 0 \rangle_{\text{be}}, \quad (3)$$

where

$$G_k = U_A^\dagger e^{i\phi_{2k} Z_\Pi} U_A e^{i\phi_{2k+1} Z_\Pi}, \quad (4a)$$

$$\langle 0 |_{\text{be}} U_A | 0 \rangle_{\text{be}} = A. \quad (4b)$$

The equation (3) and the corresponding circuit 1 compute the real polynomial $P[\phi](A)$ [3]. Here, Z_q is the Pauli Z gate acting on the qubit q . The operator Z_Π changes the sign of nonzero states of the ancillary register ‘be’ used for encoding the matrix A . The qubit q serves as the target qubit for the controlled rotations $\exp(i\phi_k Z_\Pi)$ which perform either $\exp(i\phi_k)$ or $\exp(-i\phi_k)$ transformations depending on the state of the register ‘be’.

The notation $P[\phi]$ indicates that the QSVT circuit is parameterized by a set of angles ϕ , precomputed classically. These angles determine the shape of the function $f(s)$ in Eq. (2). We are interested only in the computation of the inverse function:

$$f(s) = s^{-1}. \quad (5)$$

By approximating the function (5), QSVT inverts the original matrix (1). Let us introduce the condition number ρ_A of the matrix A :

$$\rho_A = \frac{s_{\max}}{s_{\min}} = \frac{\|A\|}{s_{\min}}. \quad (6)$$

Thus, $s_{\min} = \|A\| \rho_A^{-1}$. To avoid the singularity at $s = 0$ in Eq. (5), we approximate the inverse function (5) by the odd function [5],

$$\mathcal{F}(s) = \frac{1 - e^{-(5s\kappa_{\text{qsvt}})^2}}{s}, \quad (7)$$

which provides an ϵ -approximation of the inverse function within the interval

$$[-1, -\kappa_{\text{qsvt}}^{-1}] \cup [\kappa_{\text{qsvt}}^{-1}, 1], \quad (8)$$

with $\epsilon \leq 10^{-12}$, as one can see in Fig. 2. To guarantee that the above interval includes the entire domain of the singular values of the matrix, $\kappa_{\text{qsvt}}^{-1}$ must be less or equal than the minimal singular value of A defined according to Eq. (6):

$$\kappa_{\text{qsvt}} \geq \|A\|^{-1} \rho_A = s_{\min}^{-1}. \quad (9)$$

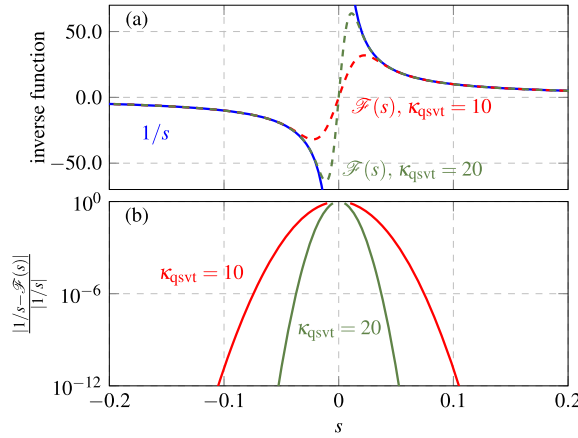


Fig. 2. (a) Plot showing the function s^{-1} (blue solid line) and the function $F(s)$ [Eq. (7)] with various κ_{qsvt} (dashed lines) constructed in the domain $s \in [-0.2, 0.2]$. (b) The relative error in approximating s^{-1} with the function $F(s)$ with various κ_{qsvt} . (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

In this case, the function $F(s)$ is ϵ -close to the inverse function (5) in the interval (8) and is finite for $s \in [-\kappa_{\text{qsvt}}^{-1}, \kappa_{\text{qsvt}}^{-1}]$, as seen from Fig. 2. It is worth mentioning that one can use other approximations of the inverse function [1,4] where the error ϵ can be varied explicitly.

To construct the inverse matrix A^{-1} by QSVT, one needs to precompute the QSVT angles ϕ setting $F(s)$ as the target function as explained in Sec. 2.2. Then, the optimized angles ϕ are used as the parameters of the QSVT circuit 1. The resulting circuit returns the quantum state

$$|\text{out}\rangle = |0\rangle_{q,\text{be}} |\psi\rangle_{\text{in}} + |\dots\rangle, \quad (10)$$

where we are interested only in the state entangled with $|0\rangle_{q,\text{be}}$,

$$|\psi\rangle_{\text{in}} = \frac{\eta_{\text{qsvt}} e^{i\phi_{\text{glob}}}}{\kappa_{\text{qsvt}}} A^{-1} |\text{init}\rangle_{\text{in}}, \quad (11)$$

where ϕ_{glob} is an unknown global phase, $\eta_{\text{qsvt}} < 1$ is a constant positive scalar that ensures smooth computation of QSVT angles by the minimization algorithm [4,21], and $|\text{init}\rangle_{\text{in}}$ is an initial state that needs to be multiplied by A^{-1} . In this work, we keep $\eta_{\text{qsvt}} = 0.125$.

2.2. Precise computation of QSVT angles

Let us assume that $F(s)$ is the target function to be computed by QSVT. In our case, we set $\eta_{\text{qsvt}} F(s) / \kappa_{\text{qsvt}}$ as the target function. To compute QSVT angles [4,21], first of all, the function $F(s)$ is approximated by a sequence of Chebyshev polynomials,

$$\tilde{F}(s) = \sum_{k=0}^{N_c} c_k T_k(s), \quad (12a)$$

$$\tilde{F}(s) = F(s) + \epsilon_{\text{qsvt}}, \quad (12b)$$

where ϵ_{qsvt} is the approximation error. The coefficients c_k can be found using the Fourier series

$$c_k \approx \frac{2 - \delta_{k0}}{2N_q} (-1)^k \sum_{j=0}^{2N_q-1} F(-\cos(j\pi/N_q)) e^{i \frac{kj\pi}{N_q}}, \quad (13)$$

where N_q should be not less than N_c , and the function $F(s)$ is assumed to be normalized such that its value is less or equal to one. Since our target function (7) is odd, it can be approximated by an odd polynomial $\tilde{F}$ with

$$N_a = N_c + 1 \quad (14)$$

coefficients where N_a is an even integer. On the other hand, for each s , the polynomial (12a) can be constructed by using the following sequence of unitaries [4]:

$$U[\alpha](s) = e^{i\alpha_0 Z} \prod_{l=1}^{N_a-1} W(s) e^{i\alpha_l Z}, \quad (15a)$$

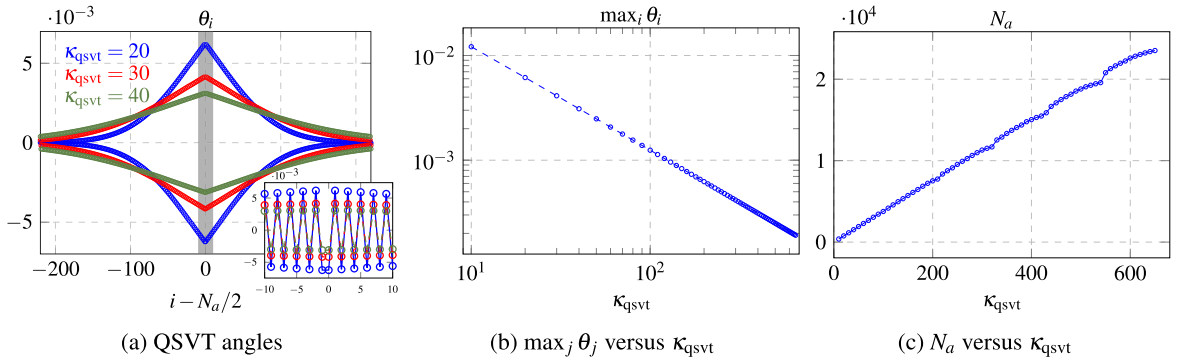


Fig. 3. (a) Plot showing the shifted QSVT angles θ_i [Eq. (19)] computed for various κ_{qsvt} using the minimization technique [4] (17) with $\epsilon_{\text{qsvt}} \approx 10^{-7}$. Note that, for the chosen κ_{qsvt} , only a subset of all angles is shown. The inner plot zooms in on θ_i within the shaded gray area. (b) The dependence of $\max_i \theta_i$ on κ_{qsvt} . (c) The dependence of the number of the QSVT angles on κ_{qsvt} .

$$W(s) = \begin{pmatrix} s & i\sqrt{1-s^2} \\ i\sqrt{1-s^2} & s \end{pmatrix}, \quad (15b)$$

$$P[\alpha](s) = \text{Re } U_{00}[\alpha](s), \quad (15c)$$

which is another representation of the QSVT circuit for some singular value s . Here, the angles α_j are shifted versions of ϕ_j :

$$\alpha_j = \begin{cases} \phi_j - \pi/4, & j = 0 \text{ and } (N_a - 1), \\ \phi_j - \pi/2, & j = 1, 2, \dots, N_a - 2. \end{cases} \quad (16)$$

The algorithm for the computation of ϕ described in [4] uses the fact that the QSVT angles ϕ satisfy the inversion symmetry, i.e. $\phi_j = \phi_{N_a-j-1}$ for $j = 0, 1, \dots, (N_a/2 - 1)$ (the same is true for α_j). Therefore, it is sufficient to compute only the first $N_a/2$ angles. The algorithm calculates these angles by minimizing the following loss function

$$L(\alpha) = \sum_{k=0}^{N_a/2-1} \left| P[\alpha](x_k^{\text{Ch}}) - \bar{F}(x_k^{\text{Ch}}) \right|^2, \quad (17)$$

where x^{Ch} are Chebyshev roots, and the ansatz $\alpha_0 = \pi/4$ and $\alpha_j = 0$ for $j = 1, 2, \dots, (N_a/2 - 1)$ is taken. If one sets the function (7) as the target function, then the minimization (17) provides such angles α that

$$P[\alpha](s) = \frac{\eta_{\text{qsvt}}}{\kappa_{\text{qsvt}}} \mathcal{F}(s) + \epsilon_{\text{qsvt}}. \quad (18)$$

It is convenient to consider QSVT angles shifted by $\pi/2$:

$$\theta_j = \phi_j - \pi/2, \quad j = 0, 1, \dots, (N_a - 1), \quad (19)$$

which are similar to α_j , except for the angles at $j = 0$ and $j = N_a - 1$. The angles θ computed by the minimization technique (17) for various κ_{qsvt} are shown in Fig. 3a. One can see that the corresponding angles ϕ are symmetric with respect to $N_a/2 - 1/2$, their values oscillate around $\pi/2$, and their amplitudes tend toward $\pi/2$ for $|j - N_a/2| \gg 1$. The maximum amplitude of θ is approximately inversely proportional to κ_{qsvt} (Fig. 3b).

The number N_a depends on the target value of the parameter κ_{qsvt} and the error ϵ_{qsvt} . More precisely, the query complexity of QSVT scales [1,4] as $\mathcal{O}(\kappa_{\text{qsvt}} \log_2(\kappa_{\text{qsvt}} \epsilon_{\text{qsvt}}^{-1}))$. For a fixed ϵ_{qsvt} , one can roughly assume that N_a grows linearly with κ_{qsvt} (Fig. 3c), becoming significant for ill-conditioned problems. Consequently, precise computation of ϕ for large condition numbers demands substantial numerical resources. Yet, in practical simulations, it is often sufficient to estimate QSVT angles rather than compute them precisely. In the next section, we propose a simple numerical algorithm to accomplish this.

3. QSVT angle estimation algorithm

Instead of directly computing ϕ for large condition numbers, we use the reference angles ϕ_{ref} provided by the minimization procedure (17) for a relatively low value κ_{ref} to estimate a set of metaparameters which describe the main features of the QSVT angles. After that, using these metaparameters, we calculate the estimated angles $\bar{\alpha}$ (and the corresponding estimated angles $\bar{\phi}$ and $\bar{\theta}$) for the target value κ_0 . Our goal is to find such $\bar{\alpha}$ that

$$P[\bar{\alpha}](s) = \frac{\eta_{\text{qsvt}}}{\kappa_0} \mathcal{F}(s) + \epsilon_{\text{appr}}, \quad (20)$$

where we set $\kappa_{\text{qsvt}} = \kappa_0$, and $\varepsilon_{\text{appr}}$ indicates the error in approximating the normalized function $F(s)$ with the polynomial $P[\bar{\alpha}](s)$ [Eq. (15)] using the estimated QSVT angles $\bar{\alpha}$. This expression is similar to that in Eq. (18), but here $\varepsilon_{\text{appr}} \geq \varepsilon_{\text{qsvt}}$ because the estimated angles $\bar{\alpha}$ usually provide worse precision than the original angles α .

To compute the metaparameters mentioned above, it is convenient to work with the angles θ . According to Fig. 3a, any two angles θ_{j-1} and θ_j have opposite signs (except the case of $j = N_a/2$ where $\theta_{j-1} = \theta_j < 0$). We assume that the envelope of the angles θ does not change with κ_0 but it is only rescaled by the κ -dependent maximum amplitude

$$\theta_{\max}(\kappa) = \max_j |\theta_j(\kappa)|. \quad (21)$$

(By the envelope of QSVT angles, we understand the change of positive and negative normalized angles $\theta_j/\theta_{\max}$ with the index j .) Therefore, to estimate $\bar{\theta}$ for an arbitrary κ_0 , one needs to find the functional dependence of $\theta_{\max}$ and N_a on the condition number, and describe the envelope of the angles. To accomplish this, we use the following technique.

- The number N_a for the target κ_0 is roughly estimated as

$$N_{a,0} = \lfloor N_{a,\text{ref}} \kappa_0 / \kappa_{\text{ref}} \rfloor, \quad (22a)$$

$$N_a = N_{a,0} + \text{mod}(N_{a,0}, 2), \quad (22b)$$

where $N_{a,\text{ref}}$ is the number of the reference angles ϕ_{ref} for a single chosen reference value κ_{ref} . Since we use the odd function (7) for approximating the inverse function, the estimated N_a should remain even [Eq. (14)]. This is ensured by Eq. (22b).

- The maximum amplitude of the shifted QSVT angles θ is approximated in the following way

$$\Theta_{\max}(\kappa) = \sum_{l=0}^{N_{\text{ampl}}-1} \frac{c_{\text{ampl},l}}{\kappa^l}, \quad (23)$$

where N_{ampl} is a chosen number of the real scalar metaparameters $c_{\text{ampl},l}$. Although Fig. 3b suggests that the optimal number of $c_{\text{ampl},l}$ is $N_{\text{ampl}} = 2$, numerical tests indicate that it is better to take $N_{\text{ampl}} = 5$. The scalars $c_{\text{ampl},l}$ are computed by minimizing the following loss function

$$L_{\text{ampl}} = \sum_{j=0}^{N_{\text{ref}}-1} |\Theta_{\max}(\kappa_{\text{ref},j}) - \theta_{\max}(\kappa_{\text{ref},j})|^2, \quad (24)$$

where N_{ref} sets of high-precision QSVT angles pre-calculated for N_{ref} values $\kappa_{\text{ref},j}$ are used.

- Then, the angles are normalized by their maximum absolute value:

$$\bar{\theta}_j = \frac{\theta_j}{\theta_{\max}}, \quad j = 0, 1, \dots, N_a - 1. \quad (25)$$

- The first half of the angles $\bar{\theta}$ is a mirror reflection of the second half (Fig. 3a). Therefore, we consider $\bar{\theta}_j$ only with $j = 0, 1, \dots, N_a/2 - 1$ which are split into

$$N_{\text{pos}} = \lfloor N_a/4 \rfloor, \quad (26a)$$

$$N_{\text{neg}} = N_a/2 - \lfloor N_a/4 \rfloor \quad (26b)$$

positive and negative angles, $\bar{\theta}_j^{\text{pos}}$ and $\bar{\theta}_j^{\text{neg}}$, correspondingly. This means that if $\text{mod}(N_a, 4) = 0$, then the number of positive and negative angles is equal.

- After that, we attribute a function $G_{\text{pos}}(r_j)$ to the normalized positive angles such that

$$G_{\text{pos}}(r_j) = \bar{\theta}_j^{\text{pos}}, \quad (27)$$

where $r_j = \Delta r j$ and $\Delta r = 1/(N_{\text{pos}} - 1)$, and assume that the function is even if one extends r from 0 to -1 . The same procedure is performed for $\bar{\theta}_j^{\text{neg}}$. After the mapping (27), we assume that the angle envelope described by the functions $G_{\text{pos}}(r_j)$ and $G_{\text{neg}}(r_j)$ does not depend on the condition number.

- Then, $G_{\text{pos}}(r_j)$ is approximated as

$$\bar{G}_{\text{pos}}(r_j) = \sum_{l=0}^{N_{\text{sh}}-1} c_{\text{sh},l}^{\text{pos}} \cos((2l) \arccos(r_j)), \quad (28)$$

where we assume that the real scalar metaparameters $c_{\text{sh},l}^{\text{pos}}$ do not depend on the condition number, and N_{sh} is a user-defined integer. The coefficients $c_{\text{sh},l}^{\text{pos}}$ are computed by minimizing the following loss function

$$L_{\text{sh}} = \sum_{j=0}^{N_{\text{pos}}-1} |\bar{G}_{\text{pos}}(r_j) - G_{\text{pos}}(r_j)|^2, \quad (29)$$

where $G_{\text{pos}}(r_j)$ is computed using $\tilde{\theta}_{\text{ref}}$ in Eqs. (25) and (27). The negative envelope $\tilde{G}_{\text{neg}}(r_j)$ is found in the same way. The resulting coefficients $c_{\text{sh},l}^{\text{neg}}$ and $c_{\text{sh},l}^{\text{pos}}$ for each l have opposite signs and may differ slightly in their absolute amplitudes. Numerical tests show that $N_{\text{sh}} = 20$ is sufficient, and further increasing N_{sh} does not significantly decrease $\varepsilon_{\text{appr}}$.

Using $N_{\text{ampl}} + 2N_{\text{sh}} + 2$ metaparameters, including the reference value κ_{ref} , the number $N_{a,\text{ref}}$ of the reference QSVT angles ϕ_{ref} , the coefficients $c_{\text{ampl},l}$, and the coefficients $c_{\text{sh},l}^{\text{pos}}$ and $c_{\text{sh},l}^{\text{neg}}$, one can now estimate the QSVT angles $\tilde{\theta}$ for the target value κ_0 .

- First of all, one estimates N_a for the chosen κ_0 using Eq. (22) and computes N_{pos} and N_{neg} using Eq. (26).
- After that, the κ -independent coefficients $c_{\text{sh},l}^{\text{pos}}$ and $c_{\text{sh},l}^{\text{neg}}$ computed in Eq. (29) are employed in Eq. (28) to build the envelope of the estimated positive and negative angles, i.e. $\tilde{\theta}_j^{\text{pos}}$ with $j = 0, 1, \dots, N_{\text{pos}} - 1$ and $\tilde{\theta}_k^{\text{neg}}$ with $k = 0, 1, \dots, N_{\text{neg}} - 1$, correspondingly.
- The order of the elements in the copies of $\tilde{\theta}^{\text{pos}}$ and $\tilde{\theta}^{\text{neg}}$ is inverted and the inverted copies are concatenated with the original $\tilde{\theta}^{\text{pos}}$ and $\tilde{\theta}^{\text{neg}}$, respectively, to recreate the entire range of the normalized estimated angles $\tilde{\theta}_j$ with $j = 0, 1, \dots, N_a - 1$.
- Using the coefficients $c_{\text{ampl},l}$ computed in Eq. (24), one finds the maximum value $\tilde{\theta}_{\text{max}}$ for the chosen κ_0 using Eq. (23).
- The final step is to renormalize $\tilde{\theta}$ by $\tilde{\theta}_{\text{max}}$,

$$\bar{\theta} = \tilde{\theta}_{\text{max}} \tilde{\theta} \quad (30)$$

which results in the estimated shifted angles $\bar{\theta}$ for the chosen κ_0 . The estimated angles $\bar{\phi}$, which are used by the QSVT circuit, can be found from $\bar{\theta}$ using Eq. (19).

The algorithm is summarized in Table 1. The code for estimating QSVT angles can be found in [22].

4. Numerical tests

To verify the technique described in Sec. 3 and compute the error $\varepsilon_{\text{appr}}$, we perform several numerical tests using the estimated QSVT angles $\bar{\phi}$. These tests include inverting the approximate inverse function (7) and a sine function, and simulating a boundary-value problem that describes the linear dynamics of EM waves in a dielectric medium. In all these tests, we use $\kappa_{\text{ref}} = 650$ to compute the envelope coefficients $c_{\text{sh},l}^{\text{pos}}$ and $c_{\text{sh},l}^{\text{neg}}$ in Eq. (29). Also, we use $N_{\text{ref}} = 65$ sets of high-precision angles with $\kappa_{\text{ref}} = 10, 20, \dots, 650$ to compute the coefficients $c_{\text{ampl},l}$ in Eq. (24). The simulated test cases and the corresponding quantum circuits can be found in [22].

4.1. Inverting $\mathcal{F}$ and sine functions

To verify that the QSVT angles are estimated correctly, we invert the normalized function (7) for various κ_{qsvt} . More precisely, we block encode the following diagonal matrix

$$A_{\mathcal{F}} = \frac{\eta_A}{\kappa_{\text{qsvt}}} \text{diag}(\mathcal{F}_0, \mathcal{F}_1, \dots, \mathcal{F}_{N_x-1}) \quad (31)$$

where $\mathcal{F}_k = \mathcal{F}(x_k)$ with x_k is a grid with $N_x = 2^{n_x}$ points satisfying Eq. (8), and $\eta_A \leq 1$ is a rescaling factor of the spectral norm of the matrix $A_{\mathcal{F}}$. The matrix $A_{\mathcal{F}}$ has singular values

$$s_k = \left| \frac{\eta_A \mathcal{F}_k}{\kappa_{\text{qsvt}}} \right|, \quad (32)$$

and its condition number ρ_A is equal to κ_{qsvt} if $\eta_A = 1$. The maximum singular value is equal to η_A . The normalization to κ_{qsvt} is necessary to make the matrix spectral norm less or equal to one which is a necessary condition for the block-encoding of $A_{\mathcal{F}}$. Then, we invert $A_{\mathcal{F}}$ by using circuit 1. There, we use the estimated QSVT angles $\bar{\phi}$ computed with the Algorithm 3 by setting $\kappa_0 = \rho_A$.

According to Eq. (11), the result of the inversion of Eq. (31) by QSVT is a state vector with the elements

$$\psi_{\text{in},k} = \frac{\eta_{\text{qsvt}} x_k}{\eta_A 2^{n_x/2}}, \quad (33)$$

where the term $2^{-n_x/2}$ appears from $|\text{init}\rangle_{\text{in}}$, and we do not take into account the global phase. After renormalizing $|\psi\rangle_{\text{in}}$ by $2^{n_x/2}/\eta_{\text{qsvt}}$, this vector encodes a linear function from $-\eta_A^{-1}$ to η_A^{-1} . The comparison between the classical and QSVT results is shown in Fig. 4a. The numerical tests show that the maximum error is localized at the singular value $s = 1$. This error can be easily avoided by setting η_A to a value less than one. In fact, it is often difficult to block-encode a matrix while keeping its spectral norm equal to one. Therefore, this localized error should not be noticeable in most QSVT simulations. According to Fig. 4c, the maximum absolute error of the QSVT computation of the normalized inverse function is around 10^{-5} for large condition numbers. The drop of the error at $\rho_A \approx 10^3$ is observed because ρ_A becomes close to the $\kappa_{\text{ref}} = 650$ used for computing the envelope coefficients of the reference QSVT angles ϕ_{ref} .

As another test, we invert the matrix

$$A_{\text{sin}} = \text{diag}(\sin \xi_k), \quad (34a)$$

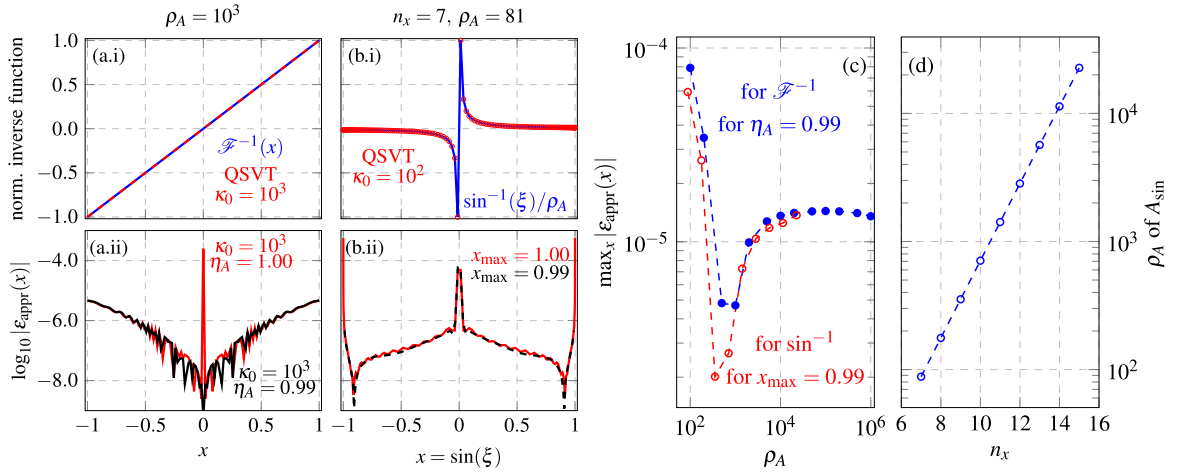


Fig. 4. (a): Plots showing comparison of the inverse function $F^{-1}(x)$ computed classically (blue line) and by QSVT (red line) for $\rho_A = 10^3$ (a.i) and the corresponding QSVT error (a.ii) with $\eta_A = 1.00$ (red line) and $\eta_A = 0.99$ (black line). (b): Comparison of the inverse function $\sin^{-1}(\xi)$ computed classically (blue line) and by the QSVT (red markers) for $n_x = 7$ and $\rho_A = 81$ (b.i) and the corresponding QSVT error (b.ii) with $x_{\max} = 1.00$ (red line) and $x_{\max} = 0.99$ (black line). (c): The maximum QSVT error for various ρ_A in the computation of the inverse functions $F^{-1}(x)$ (blue markers) and $\sin^{-1}(\xi)$ (red markers). (d): The dependence of the condition number ρ_A of the matrix $A_{\sin}$ on n_x .

$$\xi_k = -\xi_{\max} + \frac{2\xi_{\max}k}{N_x - 1}, \quad k = 0, 1, \dots, (N_x - 1), \quad (34b)$$

whose condition number ρ_A can be increased by increasing n_x as one can see from Fig. 4d. The singular values of $A_{\sin}$ are

$$s_k = |\sin \xi_k|. \quad (35)$$

If $\xi_{\max} = \pi/2$, the spectral norm of $A_{\sin}$ is one. To compute $A_{\sin}^{-1}$ using the QSVT circuit, we estimate the QSVT angles $\bar{\phi}$ by setting $\kappa_0 \geq \rho_A$. Similarly to the previous case, the maximum error of the QSVT computations with $\bar{\phi}$ is localized at $s = 1$ (Fig. 4b). By slightly decreasing $\xi_{\max}$, one can set $\|A_{\sin}\| < 1$ and, due to this, reduce the maximum error to 10^{-5} . The dependence of ϵ_{appr} on ρ_A for both $F^{-1}(x)$ and $\sin^{-1}(\xi)$ shows the same behavior as one can see from Fig. 4c.

4.2. Electromagnetic (EM) waves

Here, we consider a one-dimensional boundary-value problem describing stationary EM waves,

$$i\omega\epsilon_L E(x) + \partial_x B(x) = 0, \quad (36a)$$

$$i\omega B(x) + \partial_x E(x) = 0, \quad (36b)$$

with the outgoing boundary conditions at $x = 0$ and $x = 1$ and the source $Q = Q_0 \exp(i\omega t)$:

$$(i\omega - \partial_x)E \Big|_{x=0} = 0, \quad (37a)$$

$$(i\omega + \partial_x)B \Big|_{x=1} = Q_0, \quad (37b)$$

on a spatial grid $x = [0, 1]$ with $N_x = 2^{n_x}$ points in a medium with the following dielectric permittivity

$$\epsilon_L = \begin{cases} \epsilon_0, & x < 0.5, \\ \epsilon_1, & x > 0.5. \end{cases} \quad (38)$$

The detailed description of the system and the corresponding matrix A_{EM} encoding the EM problem are described in [8]. We use this problem as another test case for the verification of the estimated QSVT angles $\bar{\phi}$ for various κ_0 . The comparison between classical and QSVT simulations of this dynamical system in the case with $n_x = 8$ is shown in Fig. 5 where one can see the spatial distribution of the real and imaginary components of the stationary electric field E . The absolute difference between the classical and QSVT signals is around 10^{-5} . As shown in Fig. 5d, the condition number ρ_A of the matrix A_{EM} encoding the EM wave problem grows with increasing n_x , and for $n_x = 8$, it is equal to 801. The spectral norm $\|A_{\text{EM}}\|$ of the block-encoded matrix is 0.2 and practically does not depend on n_x . Hence, according to Eq. (9) and Fig. 5c, one should take $\kappa_0 \approx \|A_{\text{EM}}\|^{-1} \rho_A$ to minimize the QSVT error. With this condition satisfied, the maximum error of the QSVT computation with the estimated angles $\bar{\phi}$ is around 10^{-5} for all simulated n_x .

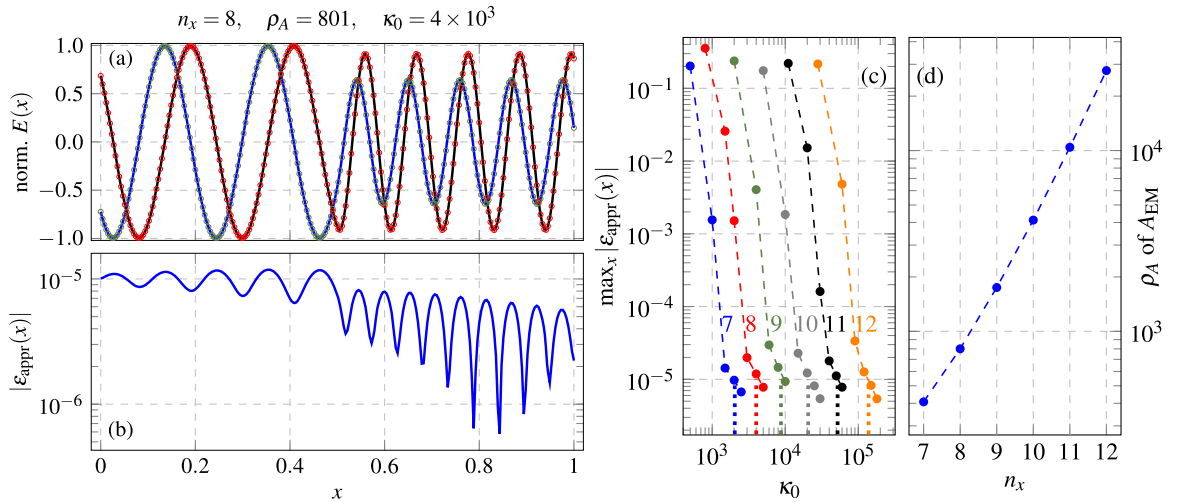


Fig. 5. (a): Plot showing the real (solid blue line) and imaginary (solid black line) components of the electric field E computed classically, and the real (green markers) and imaginary (red markers) components of E computed by the QSVT circuit 1 with $\bar{\phi}$ estimated for $\kappa_0 = 4 \times 10^3$. Here, $n_x = 8$ and $\rho_A = 801$. (b): The difference between the classical and QSVT signals. (c): The dependence of $\max_x |\epsilon_{\text{appr}}(x)|$ on the value of κ_0 used for the estimation of $\bar{\phi}$ for various n_x , which are indicated by colored integers. The colored vertical dashed lines indicate the values $5\rho_A$ for various n_x . (d): The dependence of the condition number ρ_A of the matrix A_{EM} on n_x .

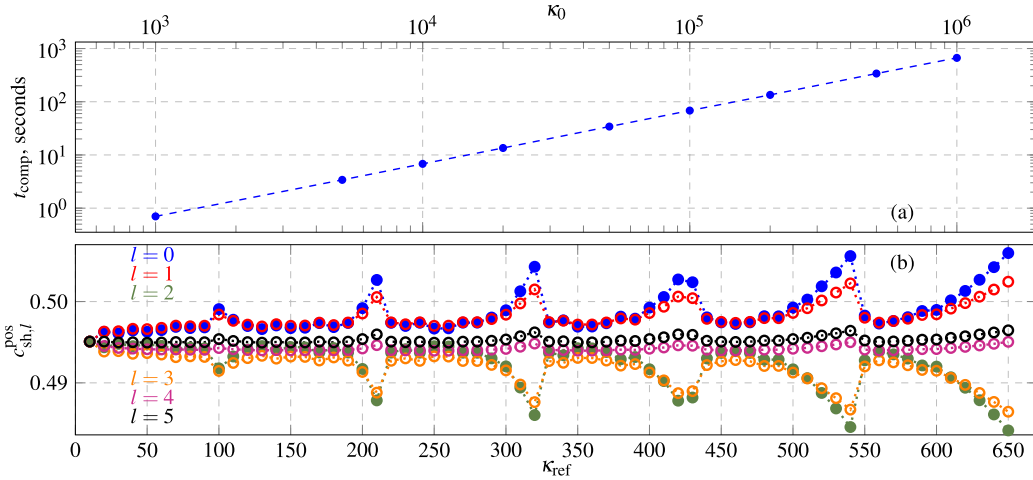


Fig. 6. (a) The dependence of the computational time t_{comp} used by the approach 3 for estimating $\bar{\phi}$ for various κ_0 . (b) The dependence of the coefficients $c_{\text{sh},l}^{\text{pos}}$ [Eq. (28)] for several l on κ_{ref} . Here, for the sake of clarity, all $c_{\text{sh},l}^{\text{pos}}$ with $l \neq 1$ are shifted towards $c_{\text{sh},1}^{\text{pos}}$ in the following way: $c_{\text{sh},l}^{\text{pos}}(\kappa_{\text{ref}}) = c_{\text{sh},1}^{\text{pos}}(\kappa_{\text{ref}}) + [c_{\text{sh},1}^{\text{pos}}(10) - c_{\text{sh},l}^{\text{pos}}(10)]$.

4.3. Discussion

According to the numerical tests discussed in the previous sections, one can expect $\epsilon_{\text{appr}} \approx 10^{-5}$ in the QSVT computations with the estimated angles $\bar{\phi}$ with $\kappa_0 \leq 10^6$. The computational time required for estimating $\bar{\phi}$ grows linearly with κ_0 as shown in Fig. 6a. To improve the precision of the proposed estimation algorithm, one can optimize N_a using the approximation (22) as an initial guess. Indeed, according to Fig. 4c, the error ϵ_{appr} increases when one computes $\bar{\phi}$ for small κ_0 . This may mean that there could be a constant factor in the scaling (22). The contribution of this term to N_a is stronger for small κ_0 . We are not interested in estimating $\bar{\phi}$ for $\kappa_0 \leq 650$ because one can relatively easily compute ϕ for low condition numbers using the high-precision algorithms [4,5]. Yet, by optimizing the estimation of N_a for low κ_0 , one can find the missing constant factor in Eq. (22) and, thus, can potentially decrease ϵ_{appr} for large κ_0 as well. Moreover, according to Fig. 3c, N_a does not grow exactly linearly with the condition number. This results in an additional error when one uses the linear scaling (22).

Another option to improve the precision of the proposed estimation technique is to take into account the dependence of the Chebyshev coefficients $c_{\text{sh},l}^{\text{pos}}$ and $c_{\text{sh},l}^{\text{neg}}$ on the condition number. In the current version of the algorithm, we compute $c_{\text{sh},l}$ for a single $\kappa_{\text{ref}} = 650$. However, $c_{\text{sh},l}$ for each particular $l = 0, 1, \dots, (N_{\text{sh}} - 1)$ slightly depends on the condition number as one can see in Fig. 6b, which shows the dependence of a few first coefficients $c_{\text{sh},l}^{\text{pos}}$ on κ_{ref} . One can reformulate the minimization problem (17) in terms of $2N_{\text{sh}}$ coefficients $c_{\text{sh},l}$ instead of N_a angles α_k . As a result, instead of optimizing N_a QSVT angles for the chosen κ_0 (e.g., $N_a \approx 4 \times 10^7$

for $\kappa_0 = 10^6$), one will need to minimize a loss function with respect to $2N_{\text{sh}} \approx 40$ real parameters $c_{\text{sh},l}$. As an option, the FFPI technique recently developed in Ref. [7] can be used to efficiently generate more data for training the parameters $c_{\text{sh},l}$ to take into account their dependence on the condition number. That said, such optimization will be more expensive than the current fast estimation technique, and it may be preferable to use the FFPI algorithm [7] to compute high-precision QSVT angles directly, although the latter has not yet been tested for such large condition numbers considered in this work.

5. Conclusions

We have proposed a new numerical technique for estimating QSVT angles for matrix inversion with large condition numbers. The algorithm is based on using a small number of metaparameters that describe the dependence of QSVT angles on the condition number. By operating with these metaparameters instead of the angles directly, we have significantly reduced the time necessary to compute the QSVT angles while maintaining relatively high precision. The computational time of this algorithm scales linearly with the target condition number. The estimated angles can be used in QSVT circuits to solve systems of linear equations with an error of 10^{-5} for condition numbers as large as 10^6 . The precision of the technique can potentially be increased by optimizing the metaparameter values for the target condition numbers.

CRedit authorship contribution statement

I. Novikau: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **I. Joseph:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this article are openly available in Ref. [22].

References

- [1] A. Gilyén, Y. Su, G.H. Low, N. Wiebe, Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics, in: *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing, STOC 2019, Association for Computing Machinery, New York, NY, USA, 2019*, pp. 193–204.
- [2] J.M. Martyn, Z.M. Rossi, A.K. Tan, I.L. Chuang, Grand unification of quantum algorithms, *PRX Quantum* 2 (2021) 040203, <https://doi.org/10.1103/PRXQuantum.2.040203>, <https://link.aps.org/doi/10.1103/PRXQuantum.2.040203>.
- [3] L. Lin, Lecture notes on quantum algorithms for scientific computation, <https://doi.org/10.48550/ARXIV.2201.08309>, <https://arxiv.org/abs/2201.08309>, 2022.
- [4] Y. Dong, X. Meng, K.B. Whaley, L. Lin, Efficient phase-factor evaluation in quantum signal processing, *Phys. Rev. A* 103 (4) (Apr 2021), <https://doi.org/10.1103/physreva.103.042419>.
- [5] L. Ying, Stable factorization for phase factors of quantum signal processing, <https://doi.org/10.48550/ARXIV.2202.02671>, <https://arxiv.org/abs/2202.02671>, 2022.
- [6] Y. Dong, L. Lin, H. Ni, J. Wang, Robust iterative method for symmetric quantum signal processing in all parameter regimes, *arXiv*, arXiv:2307.12468 [abs], 2023, <https://api.semanticscholar.org/CorpusID:260125356>.
- [7] H. Ni, L. Ying, Fast phase factor finding for quantum signal processing, *arXiv*:2410.06409, <https://arxiv.org/abs/2410.06409>, 2024.
- [8] I. Novikau, I. Dodin, E. Startsev, Simulation of linear non-Hermitian boundary-value problems with quantum singular-value transformation, *Phys. Rev. Appl.* 19 (2023) 054012, <https://doi.org/10.1103/PhysRevApplied.19.054012>, <https://link.aps.org/doi/10.1103/PhysRevApplied.19.054012>.
- [9] I. Novikau, I. Dodin, E. Startsev, Encoding of linear kinetic plasma problems in quantum circuits via data compression, *J. Plasma Phys.* 90 (4) (2024) 805900401, <https://doi.org/10.1017/S0022377824000795>.
- [10] J. Golden, D. O'Malley, H. Viswanathan, Quantum computing and preconditioners for hydrological linear systems, *Sci. Rep.* 12 (1) (2022) 22285, <https://doi.org/10.1038/s41598-022-25727-9>.
- [11] J.M. Henderson, M. Podzorova, M. Cerezo, J.K. Golden, L. Gleyzer, H.S. Viswanathan, D. O'Malley, Quantum algorithms for geologic fracture networks, *Sci. Rep.* 13 (1) (2023) 2906, <https://doi.org/10.1038/s41598-023-29643-4>.
- [12] J.M. Henderson, J. Kath, J.K. Golden, A.G. Percus, D. O'Malley, Addressing quantum's “fine print” with efficient state preparation and information extraction for quantum algorithms and geologic fracture networks, *Sci. Rep.* 14 (1) (2024) 3592, <https://doi.org/10.1038/s41598-024-52759-0>.
- [13] D.W. Berry, P.C.S. Costa, Quantum algorithm for time-dependent differential equations using Dyson series, *Quantum* 8 (2024) 1369, <https://doi.org/10.22331/q-2024-06-13-1369>.

- [14] T. Carleman, Application de la théorie des équations intégrales linéaires aux systèmes d'équations différentielles non linéaires, *Acta Math.* 59 (1932) 63–87, <https://doi.org/10.1007/BF02546499>.
- [15] J.-P. Liu, H. Kolden, H.K. Krovi, N.F. Loureiro, K. Trivisa, A.M. Childs, Efficient quantum algorithm for dissipative nonlinear differential equations, *Proc. Natl. Acad. Sci.* 118 (35) (Aug. 2021), <https://doi.org/10.1073/pnas.2026805118>.
- [16] H. Krovi, Improved quantum algorithms for linear and nonlinear differential equations, *Quantum* 7 (2023) 913, <https://doi.org/10.22331/q-2023-02-02-913>.
- [17] B.O. Koopman, Hamiltonian systems and transformations in Hilbert space, *Proc. Natl. Acad. Sci. USA* 17 (5) (1931) 315–318, <https://doi.org/10.1073/pnas.17.5.315>, <https://www.pnas.org/doi/10.1073/pnas.17.5.315>.
- [18] I. Joseph, Koopman–von Neumann approach to quantum simulation of nonlinear classical dynamics, *Phys. Rev. Res.* 2 (2020) 043102, <https://doi.org/10.1103/PhysRevResearch.2.043102>, <https://link.aps.org/doi/10.1103/PhysRevResearch.2.043102>.
- [19] I.Y. Dodin, E.A. Startsev, On applications of quantum computing to plasma simulations, *Phys. Plasmas* 28 (9) (2021) 092101, <https://doi.org/10.1063/5.0056974>.
- [20] T. Akiba, Y. Morii, K. Maruta, Carleman linearization approach for chemical kinetics integration toward quantum computation, *Sci. Rep.* 13 (1) (2023) 3935, <https://doi.org/10.1038/s41598-023-31009-9>.
- [21] Efficient phase-factor evaluation in quantum signal processing: code, <https://github.com/qsppack/QSPPACK>.
- [22] Code of the estimation algorithm, <https://github.com/QuCF/QuCF/wiki/Estimating-QSVT-angles>, 2024.